

PROJECT REPORT



PROJECT TITLE:

Alert System for Women on Vehicles

Submitted By:

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Abstract:

“Alert System for Women on Vehicles” is an embedded systems project especially designed for women safety during travel and transportation. This system is an IoT-based emergency safety device built using the ESP32 microcontroller. The system is designed to immediately alert an emergency contact when a woman traveling alone feels unsafe or is in danger.

Upon triggering the system, a buzzer activates instantly, an LCD displays the system status, GPS coordinates are acquired in real time, and an SMS containing a live Google Maps location link is automatically sent to a pre-configured emergency phone number via a GSM module. The system is compact, low cost, and can be installed on any vehicle, making it a practical solution for enhancing women's safety.

The developed prototype successfully demonstrates the capability of providing immediate assistance during emergencies by transmitting accurate location information and generating local alerts. This project can be effectively deployed in vehicles, public transportation systems, and personal safety applications to enhance women's security and support rapid emergency response.

Future enhancements may include mobile application integration, cloud-based monitoring, voice recognition, accident detection sensors, and real-time tracking through Internet of Things (IoT) platforms.

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Introduction:

Safety and security have become major concerns in today's society, particularly for women traveling alone in vehicles or unfamiliar locations. Despite technological advancements, incidents of harassment, assault, and emergencies continue to occur. Immediate communication with family members, guardians, or emergency services is crucial during such situations.

The Women Safety Alert System for Vehicles is designed to provide a rapid emergency response mechanism using modern communication and location-tracking technologies. The system integrates an ESP32 microcontroller with GPS and GSM modules to track the user's location and transmit emergency alerts. A buzzer is incorporated to generate an audible alarm, while an LCD display provides system status information.

When an emergency situation occurs, the system acquires the current GPS coordinates and sends an alert message containing the location information to predefined contacts through the GSM network. This ensures that the victim's location can be identified quickly, enabling a faster response and improving overall safety.

Problem Definition:

Despite advances in mobile technology, women in distress situations while traveling still face significant challenges:

1. **Increasing Safety Concerns for Women:** Women's safety has become a major concern in society. Many women face risks while traveling alone in vehicles, especially during late hours. Emergency situations require immediate communication and assistance.
2. **Delay in Emergency Response:** During emergencies, it may be difficult for victims to contact family members or authorities quickly. Delayed communication can increase the severity of the situation. A rapid alert mechanism is essential to ensure timely help.
3. **Dependence on Smartphones:** Most existing safety solutions are mobile application-based. These applications require smartphones to be accessible during emergencies. Victims may not always be able to operate their phones in critical situations.
4. **Requirement of Internet Connectivity:** Many safety applications depend on internet services. Poor network coverage or lack of internet access can prevent emergency alerts from being sent. This reduces the reliability of existing systems.
5. **Difficulty in Sharing Accurate Location:** During emergencies, victims may not know their exact location. Providing inaccurate location details can delay rescue operations. A system capable of automatically sharing precise coordinates is required.

Objectives:

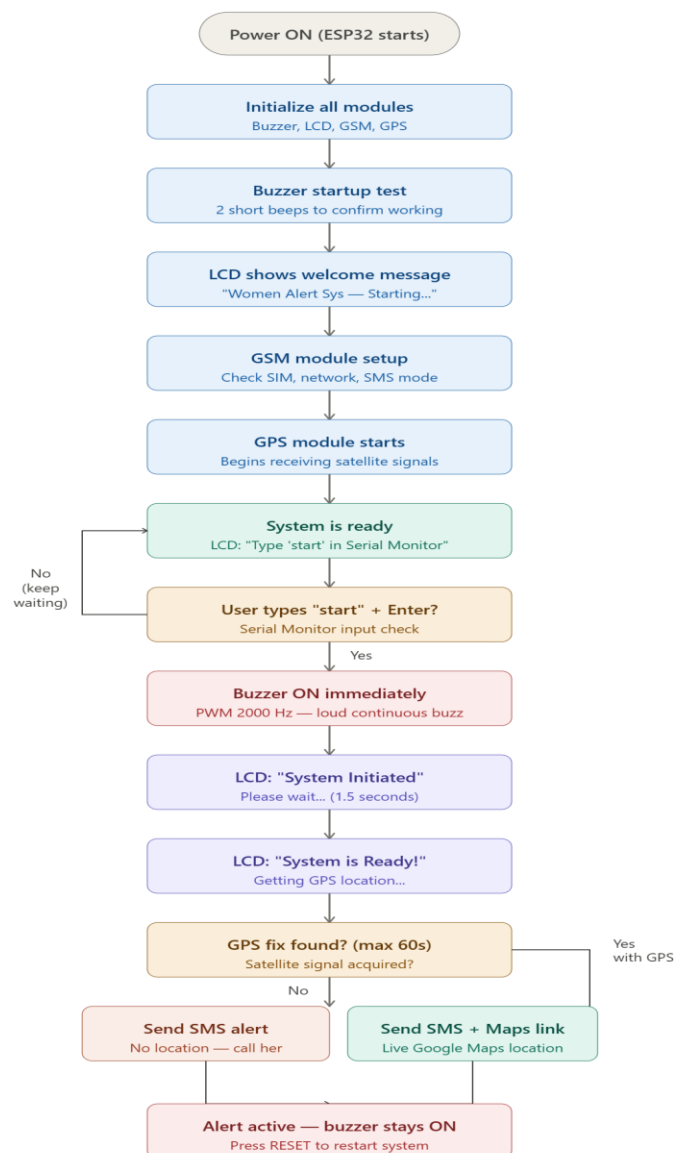
The main objective of the Alert System for women on vehicles project is to design a women safety system while travelling and transportation when she feels unsafe or in danger. The other specific objectives as follows:

1. To design and implement a low-cost IoT-based emergency alert system for women's safety.
2. To integrate GPS technology for real-time live location tracking.
3. To use GSM communication to send automatic SMS alerts to emergency contacts.
4. To provide immediate audio alert through a buzzer upon system activation.
5. To display real-time system status on an LCD screen.
6. To ensure the system works reliably on vehicles in outdoor environments.
7. To create a system that can be easily operated without requiring technical knowledge.
8. To showcase the practical implementation of embedded systems and wireless communication technologies in solving real-world safety problems.

Methodology/Procedure:

The methodology of the proposed Women Safety Alert System involves the integration of ESP32, GPS, GSM, LCD, and buzzer modules to provide a reliable emergency alert mechanism. The system is designed to obtain the user's current location, transmit emergency notifications, and generate audible alerts during critical situations. The implementation process consists of hardware integration, software development, communication establishment, location acquisition, message transmission, and alert generation.

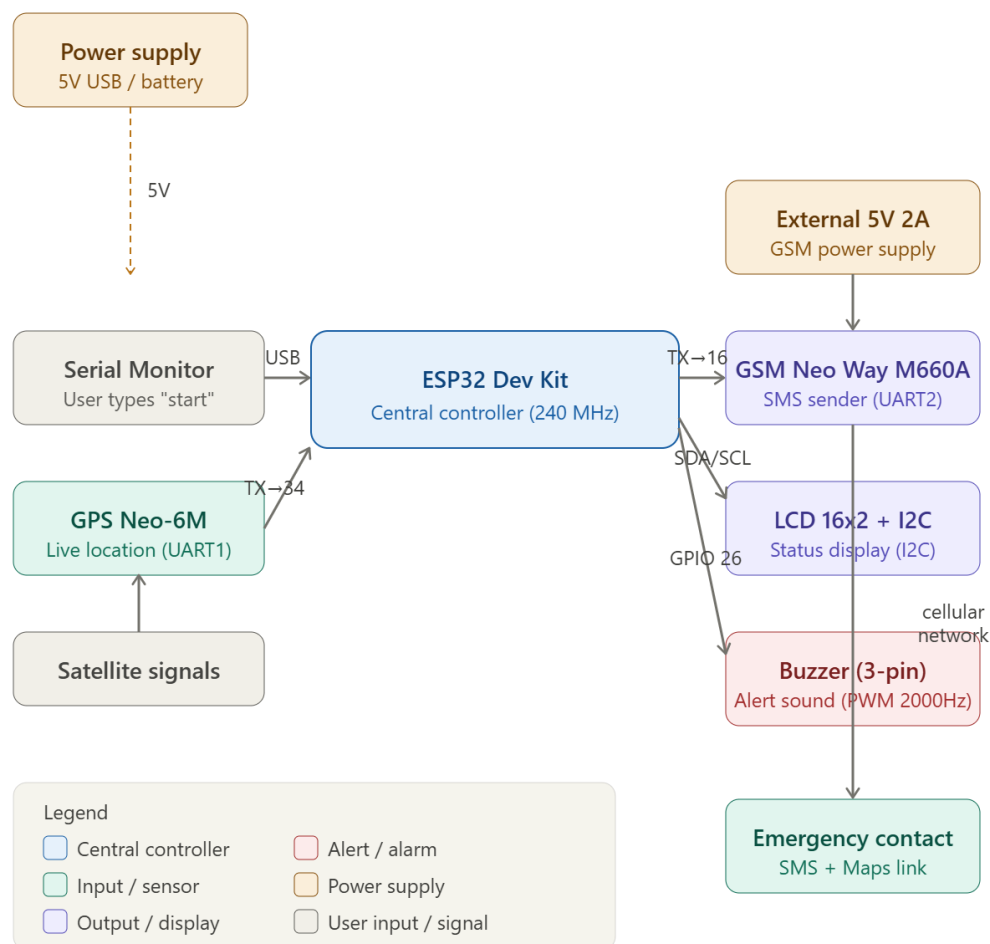
Flowchart: Alert system for Women on vehicles



System Architecture:

This women alert system is designed using an ESP32 as the central controller connected to a GPS module, GSM module, LCD display, and buzzer. The GPS module provides location coordinates, which are processed by the ESP32. During emergencies, the ESP32 sends the location through the GSM module as an SMS to predefined contacts. Simultaneously, the LCD displays system status and the buzzer generates an audible alert. The system continues operation until a valid STOP command is received.

Block diagram:



Component Specifications:

S. No	Component Name	Quantity	Specifications/Description	Purpose
1.	Microcontroller	1	ESP32 Dev kit	Brain of the microcontroller
2.	GPS Module	1	Neo-6M	Acquires live location
3.	GSM Module	1	Neo way M660A	Sends emergency SMS to phone number
4.	LCD Display	1	16x2 Lcd display	Shows the system status
5.	Buzzer	1	3-pin active buzzer	Produces continuous loud alert

S.No	Component Name	Quantity	Specifications/Description	Purpose
6.	Breadboard	1	830 connection point	Connecting the components
7.	Jumper wires	20-25	Male to male, male to female, female to female	Connecting the components
8.	External power	1	5V 2A adapter	Supplies power to GSM

Connection Guide:

Component	ESP32 pin	Type of connection	Description
Lcd I2C	SDA→GPIO21 SCL→GPIO22	I2c Communication	Transfers data between Esp32 and Lcd display
Lcd I2c	Vcc→Vin(5V)	Power supply	Supplies power to Lcd display
Lcd I2c	GND→GND	Ground connection	Common ground for reference
GPS Module	Tx→GPIO2 Rx→GPIO4	UART Communication	Data transfers happens through these
GPS Module	Vcc→3.3v	Power supply	Supplies operating voltage to GPS Module
GPS Module	GND→GND	Ground connection	Common ground for reference
GSM Module	Tx→GPIO27 Rx→GPIO26	UART Communication	Transfers data between ESP32 and GSM Module
GSM Module	12v adaper	Power supply	Provides power to GSM Module
GSM Module	GND→GND	Ground connection	Common ground between ESP32 and GSM Module
Buzzer (signal pin)	I/O→GPIO19	Digital output	Activates buzzer during emergency
Buzzer	Vcc→3.3v	Power supply	Supplies operating voltage to buzzer
Buzzer	GND→GND	Ground connection	Common ground for reference
ESP32 USB port	USB cable	Programming & power	Used for uploading and powering ESP32 during development

Hardware design and Components:

The hardware design of the Women Safety Alert System is centered around the ESP32 microcontroller, which acts as the main control unit. The system integrates a GPS module for location tracking, a GSM module for emergency communication, an LCD display for status monitoring, and a buzzer for generating alerts. The major components and their uses are as follows:

1. ESP32 Development Board:

The ESP32 is a powerful microcontroller developed by Espressif Systems. It acts as the brain of the entire project and controls all system operations.

2. GPS Module (NEO-6M):

The Global Positioning System (GPS) module is used to determine the geographical location of the vehicle. It receives signals from GPS satellites and calculates the latitude and longitude coordinates.

3. GSM Module (Neoway M660A):

The GSM module enables wireless communication using the cellular network. It sends and receives SMS messages using a SIM card.

4. 16×2 LCD Display:

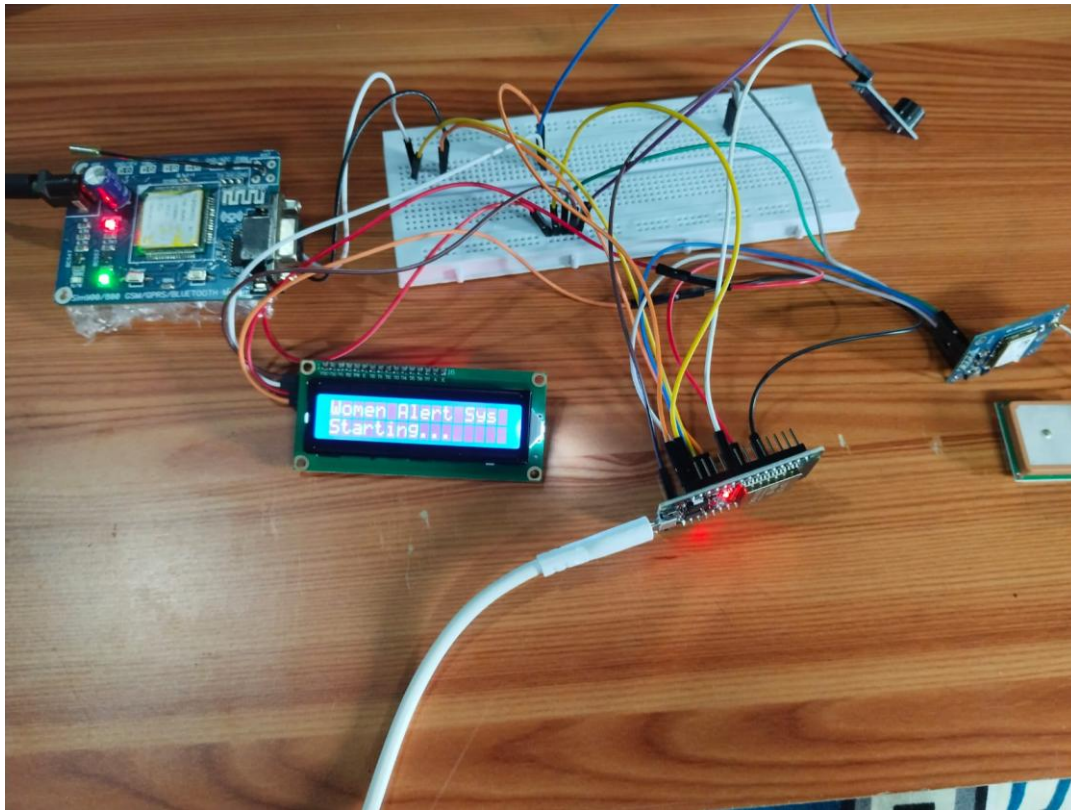
The LCD display is used to provide visual feedback to the user. It displays system status, GPS information, and emergency notifications.

5. I2C Module:

The I2C module is attached to the back of the LCD display. It reduces the number of wires required for LCD communication.

6. Active Buzzer: The buzzer is an electronic sound-producing device used to generate an audible alarm during emergencies.

Circuit Diagram:



Working Principle:

The system begins operation when power is supplied to the ESP32. The GPS, GSM, LCD, and buzzer modules are initialized. The LCD displays a ready message indicating successful initialization.

The GPS module continuously receives satellite signals and provides location information. During an emergency situation, the system obtains the current GPS coordinates and generates an emergency SMS containing the location information.

The GSM module transmits this message to predefined emergency contacts. Simultaneously, the buzzer is activated to alert nearby individuals. The LCD displays the current status of the emergency operation.

The buzzer continues operating until a predefined STOP message is received from the authorized phone number or the specified time limit is reached.

Results and Discussions:

The developed Women Safety Alert System successfully demonstrated emergency communication functionality. GPS module successfully acquired location coordinates. GSM module successfully transmitted SMS alerts. LCD displayed system status correctly. Buzzer generated audible alerts. Emergency location was accurately transmitted.

Sample SMS:

EMERGENCY ALERT!
Need Immediate Assistance.
Live location:
<https://maps.google.com/?q=15.898696,80.662437> Please reach immediately!

Conclusion:

The Women Safety Alert System for Vehicles was successfully designed and implemented using ESP32, GPS, GSM, LCD, and buzzer modules. The primary objective of the project was to provide a reliable emergency assistance mechanism capable of transmitting location information and generating alerts during critical situations.

The GPS module accurately obtains the current geographical coordinates of the user, while the GSM module sends emergency SMS messages containing location details to predefined contacts. The LCD display provides real-time system status information, and the buzzer generates an audible alert to attract the attention of nearby individuals. The integration of these components results in a dedicated hardware-based safety solution that operates independently of internet connectivity.

The developed system offers several advantages such as low cost, ease of implementation, portability, and reliable communication.

Future Scope:

The proposed Women Safety Alert System can be further enhanced by integrating advanced technologies to improve its functionality and reliability. A mobile application can be developed to provide real-time tracking and instant notifications to family members or emergency contacts. IoT and cloud-based platforms can also be incorporated to enable continuous monitoring and storage of location data.

In the future, additional features such as voice command activation, automatic accident detection, and emergency calling can be implemented to make the system more efficient and user-friendly. Camera modules and artificial intelligence techniques may also be integrated to provide better security, real-time monitoring, and faster emergency response. These improvements can make the system more intelligent, reliable, and suitable for practical deployment in modern vehicles.

References:

1.Espressif Systems, Esp32 Arduino Documentation

<https://docs.espressif.com/projects/arduino-esp32/en/latest/>

2.Arduino, Arduino Ide software

<https://www.arduino.cc/en/software>

3. John Rickman, Liquid crystal I2c Library

https://github.com/johnrickman/LiquidCrystal_I2C

4. Arduino, wire library reference

<https://www.arduino.cc/reference/en/language/functions/communication/wire/>

Appendix:

//-----**Libraries**-----

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <TinyGPS++.h>

#include <HardwareSerial.h>

//-----**Pin Definitions**-----

#define BUZZER 26

#define GSM_RX 16

#define GSM_TX 17

#define GPS_RX 4

#define GPS_TX 2

//-----**Buzzer PWM** -----

#define BUZZER_FREQUENCY 2000

#define BUZZER_RESOLUTION 8

#define BUZZER_DUTY 128

String phoneNumber = "+91xxxxxxxxxx";

#define LCD_ADDRESS 0x27

LiquidCrystal_I2C lcd(LCD_ADDRESS, 16, 2);

```
TinyGPSPlus gps;
```

```
HardwareSerial gsmModule(2);
```

```
HardwareSerial gpsModule(1);
```

```
//-----GLOBAL VARIABLES-----
```

```
String typedText = "";
```

```
bool alertDone = false;
```

```
void showLCD(String line1, String line2) {
```

```
    lcd.clear();
```

```
    delay(10);
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print(line1);
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print(line2);
```

```
}
```

```
//-----Buzzer Functions-----
```

```
void buzzerON() {
```

```
    ledcWrite(BUZZER, BUZZER_DUTY);
```

```
}
```

```
void buzzerOFF() {
```

```
    ledcWrite(BUZZER, 0);
```

```
}
```

```
void buzzerBeep(int times) {
```

```
    for (int i = 0; i < times; i++) {
```

```

    buzzerON();

    delay(300);

    buzzerOFF();

    delay(200);

}

}

//-----SETUP FUNCTION-----

void setup() {

    Serial.begin(115200);

    delay(1000);

    Serial.println("=====");

    Serial.println("  WOMEN ALERT SYSTEM STARTING");

    Serial.println("=====");

    //-----BUZZER SETUP-----

    ledcAttach(BUZZER, BUZZER_FREQUENCY, BUZZER_RESOLUTION);

    buzzerOFF();

    Serial.println("[BUZZER] Testing...");

    buzzerBeep(2);

    Serial.println("[BUZZER] Test done. Did you hear 2 beeps?");

    Serial.println("[LCD] Starting...");

    Wire.begin(21, 22);

    Wire.setClock(100000);

    delay(200);

```

```
lcd.init();  
delay(100);  
lcd.init();  
delay(100);  
lcd.backlight();  
delay(200);  
showLCD("Women Alert Sys", "Starting...");  
Serial.println("[LCD] Started.");  
delay(2000);  
Serial.println("[GSM] Starting...");  
gsmModule.begin(9600, SERIAL_8N1, GSM_RX, GSM_TX);  
delay(3000);  
gsmModule.println("AT");  
delay(1000);  
String gsmResponse = readGSM(2000);  
Serial.println("[GSM] Response: " + gsmResponse);  
if (gsmResponse.indexOf("OK") != -1) {  
    Serial.println("[GSM] GSM is working!");  
} else {  
    Serial.println("[GSM] WARNING: GSM not responding. Check power and  
wiring.");  
}  
gsmModule.println("ATE0");  
delay(500);
```

```
readGSM(500);

gsmModule.println("AT+CPIN?");

delay(1000);

String simStatus = readGSM(2000);

Serial.println("[GSM] SIM: " + simStatus);

if (simStatus.indexOf("READY") != -1) {

    Serial.println("[GSM] SIM card OK!");

} else {

    Serial.println("[GSM] WARNING: SIM problem. Check SIM card.");

}

gsmModule.println("AT+CREG?");

delay(1000);

String netStatus = readGSM(2000);

Serial.println("[GSM] Network: " + netStatus);

gsmModule.println("AT+CMGF=1");

delay(500);

readGSM(500);

Serial.println("[GSM] Setup complete.");

Serial.println("[GPS] Starting...");

gpsModule.begin(9600, SERIAL_8N1, GPS_RX, GPS_TX);

Serial.println("[GPS] Started. GPS needs open sky to get signal.");

showLCD("Type 'start' in", "Serial Monitor");
```

```

Serial.println("");

Serial.println("=====");

Serial.println("    SYSTEM IS READY    ");

Serial.println(" Type start + press Enter ");

Serial.println("=====");

}

//-----MAIN LOOP FUNCTION-----

void loop() {

    while (gpsModule.available()) {

        gps.encode(gpsModule.read());

    }

    while (Serial.available()) {

        char letter = (char)Serial.read();

        if (letter == '\n' || letter == '\r') {

            typedText.trim();

            typedText.toLowerCase();

            if (typedText == "start" && alertDone == false) {

                alertDone = true;

                runAlertSystem();

            } else if (alertDone == false) {

                Serial.println("Type start and press Enter to trigger alert.");

            }

        }

    }

}

```

```
        typedText = "";
    } else {
        typedText += letter;
    }
}
}

//-----Run Alert System-----
```

```
void runAlertSystem() {
    Serial.println("");
    Serial.println("==== ALERT TRIGGERED! ====");
    buzzerON();
    Serial.println("[BUZZER] Buzzer is ON!");
    showLCD("System Initiated", "Please wait...");
    Serial.println("[LCD] System Initiated");
    delay(1500);

    showLCD("System is Ready!", "Getting GPS...");
    Serial.println("[LCD] System is Ready!");
    Serial.println("[GPS] Getting live location...");
    double myLatitude = 0.0;
    double myLongitude = 0.0;
```

```
bool gotLocation = getGPSCoordinates(myLatitude, myLongitude);  
if (gotLocation == true) {  
    String mapLink = "https://maps.google.com/?q=" + String(myLatitude, 6)  
        + "," + String(myLongitude, 6);  
    Serial.println("[GPS] Location: " + mapLink);  
    showLCD("Sending SMS...", "Please wait...");  
    String smsText = "EMERGENCY ALERT!\n Need Immediate Assistance.\n Live location: \n"  
        + mapLink  
        + " Please reach immediately!";  
    sendSMS(smsText);  
    showLCD("Alert SMS Sent!", "Stay Safe!");  
    Serial.println("[SMS] Alert sent successfully!");  
} else {  
    Serial.println("[GPS] Failed. Sending SMS without location.");  
    showLCD("No GPS Signal", "Sending SMS...");  
    sendSMS("EMERGENCY ALERT! She needs HELP! GPS not available.  
Please call her immediately!");  
    showLCD("Alert Sent!", "(No GPS)");  
}  
delay(2000);
```



```

    }

    int secondsPassed = (millis() - startTime) / 1000;

    if (secondsPassed % 5 == 0) {
        Serial.println("[GPS] Searching... " + String(secondsPassed) + "s");
    }

    showLCD("Getting GPS...", "Wait: " + String(secondsPassed) + "s");

    delay(1000);
}

Serial.println("[GPS] Timed out after 60 seconds.");

return false;
}

//-----Send SMS-----

void sendSMS(String message) {
    Serial.println("[SMS] Sending to: " + phoneNumber);
    Serial.println("[SMS] Message: " + message);
    gsmModule.println("AT+CMGF=1");
    delay(1000);

    readGSM(1000);

    gsmModule.print("AT+CMGS=\"");
    gsmModule.print(phoneNumber);
    gsmModule.println("\");

```

```
delay(2000);

String prompt = readGSM(2000);

Serial.println("[SMS] Prompt: " + prompt);

if (prompt.indexOf(">") != -1) {

    gsmModule.print(message);

    delay(500);

    gsmModule.write(26);

    delay(5000);

String result = readGSM(5000);

    Serial.println("[SMS] Result: " + result);

    if (result.indexOf("+CMGS") != -1) {

        Serial.println("[SMS] SMS SENT SUCCESSFULLY!");

    } else {

        Serial.println("[SMS] May have failed. Check SIM balance and signal.");

    }

} else {

    Serial.println("[SMS] ERROR: No > prompt. Check GSM power and SIM.");

}

}

String readGSM(unsigned long waitMs) {

    String response = "";
```

```
unsigned long start = millis();

while (millis() - start < waitMs) {

    while (gsmModule.available()) {

        response += (char)gsmModule.read();

    }

}

response.trim();

return response;

}
```

Simulation Video Link:

https://drive.google.com/file/d/1fPchJ5UbtBN5W1B01fJFb_-i7rjp1zq0/view?usp=drivesdk

*** /THE END/***